Code No: 156AM **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, August - 2022
CONCURRENT PROGRAMMING
(Computer Science and Engineering)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a)	Explain Lamport's Bakery Algorithm with example.	
b)	Define mutual exclusion and critical section with examples.	[7+8]
2.a)	Discuss briefly about producer and consumer problem.	
b)	What are the Harsh Realities of Parallelization?	[8+7]
3.a)	Discuss the java memory model in detail.	
b)	What are the dependent progress conditions?	[8+7]
4.a)	What is the importance of sequential objects in concurrent objects?	
b)	Write algorithm on a lock-based FIFO queue.	[7+8]
5.a)	Define States and Valence with examples.	
b)	Explain importance of decide() method in Consensus Numbers.	[7+8]
6.a)	Explain in brief about working of spin locks.	
b)	What is the mechanism of test and set locks?	[8+7]
7.a)	Compare Lazy Synchronization and Non-Blocking Synchronization.	
b)	Write the Role of Locking in linked lists.	[8+7]
8.a)	Write about a Bounded Partial Queue and an Unbounded Total Queue.	
b)	Explain in brief about Naive Synchronous Queue.	[8+7]

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